

Project: CertifyPro

Professional Certification & Examination Management System

Domain: Education & Learning

1. Introduction

CertifyPro is a web-based professional certification and examination management system designed for professional bodies, examination boards, certification authorities, and regulatory institutes to manage candidate registrations, examination scheduling, test centre operations, result processing, certificate issuance, and certification renewal lifecycle management. The system enables candidates to register and schedule examinations, test centre administrators to manage exam room assignments and invigilator deployment, examination controllers to process answer scripts and publish results, and governing bodies to manage certification validity and renewal workflows.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Candidate:** Registers for certification programs, schedules examinations, and tracks result and certificate status
- **Test Centre Administrator:** Manages exam room setup, invigilator assignments, and attendance sheets
- **Examination Controller:** Oversees question paper management, result processing, and publication workflows
- **Evaluator / Examiner:** Grades answer scripts and submits marks for assigned candidate batches
- **Certification Officer:** Issues certificates, manages renewal applications, and tracks validity periods
- **Programme Admin:** Configures certification programs, exam syllabi, eligibility rules, and fee structures

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.

- **2.2 Candidate Registration & Eligibility Management:** Manages candidate profiles, qualification verification, and exam eligibility.
- **2.3 Examination Scheduling & Test Centre Management:** Manages exam windows, centre capacity, and candidate seat allocation.
- **2.4 Question Bank & Paper Management:** Manages question repositories, paper sets, and secure distribution workflows.
- **2.5 Result Processing & Grade Publication:** Manages answer script evaluation, marks entry, moderation, and result publication.
- **2.6 Certificate Issuance & Renewal Management:** Issues certificates, tracks validity periods, and manages renewal applications.
- **2.7 Examination Analytics & Reporting:** Provides dashboards for pass rates, centre performance, and certification statistics.
- **2.8 Notifications & Alerts:** Exam schedule confirmations, result availability alerts, certificate expiry reminders, and renewal deadline notifications.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Candidate and examining body staff registration with role-based access control
- Secure session management and full audit logging for all examination-critical actions

Entities

- **User**
 - UserID
 - Name
 - Role
(Candidate/CentreAdmin/ExamController/Evaluator/CertificationOfficer/Admin)
 - Email

- Phone
- Status (Active/Inactive/Suspended)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Module
 - EntityID
 - Timestamp

4.2 Candidate Registration & Eligibility Management Module

Features

- Register candidates with educational qualifications and professional experience details
- Validate eligibility against program prerequisites before allowing exam registration

Entities

- **Candidate**
 - CandidateID
 - UserID
 - Name
 - DateOfBirth
 - Gender
 - Email
 - Phone
 - Address
 - HighestQualification
 - ProfessionalExperience
 - EmployerName
 - Status (Active/Suspended/Debarred)

- **ProgramEnrolment**
 - EnrolmentID
 - CandidateID
 - ProgramID
 - EnrolmentDate
 - EligibilityStatus (Eligible/Ineligible/PendingVerification)
 - AttemptsUsed
 - MaxAttempts
 - Status (Active/Completed/Lapsed/Withdrawn)
- **CertificationProgram**
 - ProgramID
 - ProgramName
 - Body
 - Level (Foundation/Associate/Professional/Fellow)
 - EligibilityCriteria
 - ExamFee
 - ValidityYears
 - MaxAttempts
 - Status (Active/Discontinued)

4.3 Examination Scheduling & Test Centre Management Module

Features

- Configure examination windows and allocate candidates to test centres with seat management
- Manage invigilator assignments and generate hall tickets and attendance registers

Entities

- **ExamWindow**
 - WindowID
 - ProgramID

- ExamName
- StartDate
- EndDate
- RegistrationDeadline
- ResultDate
- Status (Upcoming/Open/Closed/ResultsPublished)
- **TestCentre**
 - CentreID
 - CentreName
 - City
 - Address
 - Capacity
 - ContactPerson
 - Status (Active/Inactive/Blacklisted)
- **SeatAllocation**
 - AllocationID
 - CandidateID
 - WindowID
 - CentreID
 - RoomNumber
 - SeatNumber
 - HallTicketNumber
 - Status (Allocated/Confirmed/Cancelled/NoShow)

4.4 Question Bank & Paper Management Module

Features

- Maintain a structured question bank with topic tags, difficulty levels, and marks weightage
- Assemble and securely version question papers with controlled access

Entities

- **Question**
 - QuestionID
 - ProgramID
 - TopicTag
 - Difficulty (Easy/Medium/Hard)
 - QuestionText
 - Type (MCQ/Descriptive/CaseStudy/Practical)
 - Marks
 - CreatedByID
 - Status (Active/Retired/UnderReview)
- **QuestionPaper**
 - PaperID
 - WindowID
 - ProgramID
 - PaperCode
 - TotalMarks
 - Duration
 - InstructionsRef
 - CreatedByID
 - Status (Draft/Finalised/Distributed/Archived)
- **PaperQuestion**
 - PaperQuestionID
 - PaperID
 - QuestionID
 - SequenceOrder
 - MarksAllocated

4.5 Result Processing & Grade Publication Module

Features

- Assign answer scripts to evaluators and capture marks with dual marking for quality control
- Apply moderation rules, compute final scores, and publish results with re-evaluation workflow

Entities

- **ScriptAllocation**
 - ScriptID
 - AllocationID
 - EvaluatorID
 - PaperID
 - AllocationDate
 - Status (Assigned/UnderEvaluation/MarksSubmitted)
- **MarksEntry**
 - MarksID
 - ScriptID
 - EvaluatorID
 - MarksAwarded
 - EntryDate
 - VerifiedByID
 - Status (Draft/Submitted/Verified/Moderated)
- **CandidateResult**
 - ResultID
 - CandidateID
 - WindowID
 - ProgramID
 - TotalMarks
 - MarksObtained

- Percentage
- Grade
- Outcome (Pass/Fail/Absent/Withheld)
- PublishedDate
- Status (Draft/Published/UnderReEvaluation/Revised)

4.6 Certificate Issuance & Renewal Management Module

Features

- Issue digital and physical certificates upon successful programme completion
- Manage certificate validity, renewal applications, and CPD-based renewal eligibility

Entities

- **Certificate**
 - CertificateID
 - CandidateID
 - ProgramID
 - CertificateNumber
 - IssuedDate
 - ValidUntil
 - IssuedByID
 - Status (Valid/Expired/Revoked/Suspended)
- **RenewalApplication**
 - RenewalID
 - CertificateID
 - CandidateID
 - CPDPointsSubmitted
 - ApplicationDate
 - ReviewedByID
 - NewValidUntil

- Status (Submitted/UnderReview/Approved/Rejected)

4.7 Examination Analytics & Reporting Module

Features

- Pass rate, grade distribution, and attempt utilisation dashboards by programme and exam window
- Test centre performance, no-show rates, and certification renewal compliance reports

Entities

- **ExaminationReport**
 - ReportID
 - Scope (Program/Window/Centre/Period)
 - Metrics (RegisteredCandidates, AttendanceRate, PassRate, AvgScore, CertificatesIssued, RenewalComplianceRate, CentreCapacityUtilisation)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Exam schedule confirmation, hall ticket availability, and result publication alerts for candidates
- Renewal deadline reminders, certificate expiry warnings, and evaluator assignment notifications

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Registration/Exam/Result/Certificate/Renewal)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Module, EntityID, Timestamp)
- **CertificationProgram**(ProgramID, ProgramName, Body, Level, EligibilityCriteria, ExamFee, ValidityYears, MaxAttempts, Status)
- **Candidate**(CandidateID, UserID, Name, DateOfBirth, Gender, Email, Phone, Address, HighestQualification, ProfessionalExperience, EmployerName, Status)
- **ProgramEnrolment**(EnrolmentID, CandidateID, ProgramID, EnrolmentDate, EligibilityStatus, AttemptsUsed, MaxAttempts, Status)
- **ExamWindow**(WindowID, ProgramID, ExamName, StartDate, EndDate, RegistrationDeadline, ResultDate, Status)
- **TestCentre**(CentreID, CentreName, City, Address, Capacity, ContactPerson, Status)
- **SeatAllocation**(AllocationID, CandidateID, WindowID, CentreID, RoomNumber, SeatNumber, HallTicketNumber, Status)
- **Question**(QuestionID, ProgramID, TopicTag, Difficulty, QuestionText, Type, Marks, CreatedByID, Status)
- **QuestionPaper**(PaperID, WindowID, ProgramID, PaperCode, TotalMarks, Duration, InstructionsRef, CreatedByID, Status)
- **PaperQuestion**(PaperQuestionID, PaperID, QuestionID, SequenceOrder, MarksAllocated)
- **ScriptAllocation**(ScriptID, AllocationID, EvaluatorID, PaperID, AllocationDate, Status)
- **MarksEntry**(MarksID, ScriptID, EvaluatorID, MarksAwarded, EntryDate, VerifiedByID, Status)
- **CandidateResult**(ResultID, CandidateID, WindowID, ProgramID, TotalMarks, MarksObtained, Percentage, Grade, Outcome, PublishedDate, Status)
- **Certificate**(CertificateID, CandidateID, ProgramID, CertificateNumber, IssuedDate, ValidUntil, IssuedByID, Status)

- **RenewalApplication**(RenewalID, CertificateID, CandidateID, CPDPPointsSubmitted, ApplicationDate, ReviewedByID, NewValidUntil, Status)
- **ExaminationReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Candidate Portal:** Programme catalog, exam registration, seat allocation, hall ticket download, result checker, certificate wallet
- **Test Centre Admin Console:** Candidate attendance sheet, room assignment board, invigilator deployment form
- **Examination Controller Dashboard:** Paper finalisation tracker, script allocation board, moderation workflow, result publication panel
- **Evaluator Workbench:** Assigned script list, marks entry form, dual marking comparison view
- **Certification Officer Console:** Certificate issuance queue, renewal application reviewer, validity tracker
- **Admin Console:** Programme builder, eligibility rule configurator, fee structure, window and centre management

8. Non-Functional Requirements

- **Performance:** Support 50,000+ concurrent candidate users during exam registration windows and result publication events
- **Security:** Candidate PII protection, question paper access control, RBAC, and tamper-proof audit trails for marks entries
- **Scalability:** Multi-programme and multi-centre deployment for national and international examination bodies
- **Availability:** 99.9% uptime with result publication and certificate issuance modules prioritised during peak events
- **Maintainability:** Configurable eligibility rules, grading scales, moderation policies, and renewal criteria without code changes
- **Observability:** Real-time registration dashboards, marks entry progress tracking, and centre utilisation monitoring

9. Assumptions & Constraints

- Phase 1 excludes online proctored exam delivery; all exams are conducted at registered physical test centres
- Question paper distribution is managed via secure internal access; physical paper printing workflows are out of scope
- All notifications are in-app only; SMS and email delivery integrations for candidates are deferred to Phase 2
- CPD points for renewal are self-declared by candidates; third-party CPD tracking integration is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling